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Dynamic generation of order entry fields on a trading interface

Abstract

Some embodiments of the invention provide systems and methods for displaying market data for a tradable object which are configured to generate order entry fields for display responsive to user input wherein the generated fields are automatically associated with a price for the tradable object to facilitate the timely placement of a trade.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/880,372, filed May 21, 2020, which is a continuation of U.S. patent application Ser. No. 16/008,642, filed Jun. 14, 2018, now U.S. Pat. No. 10,699,339, which is a continuation of U.S. patent application Ser. No. 15/679,865, filed Aug. 17, 2017, now U.S. Pat. No. 10,026,127, which is a continuation of U.S. patent application Ser. No. 13/935,208, filed Jul. 3, 2013, now U.S. Pat. No. 9,767,515, the contents of each of which are fully incorporated herein by reference for all purposes.

BACKGROUND

- (1) An electronic trading system generally includes a trading device in communication with an electronic exchange. The electronic exchange sends information about a market, such as prices and quantities, to the trading device. The trading device sends messages, such as messages related to orders, to the electronic exchange. The electronic exchange attempts to match quantity of an order with quantity of one or more contra-side orders.
 - (2) It should be understood that the trading device may be any computing device, such as a personal computer, laptop computer, hand-held computer, tablet computer, smart phone or some other device that includes a display, user interface and includes hardware for communication with the electronic exchange.
 - (3) Traders often use automated or semi-automated trading tools, collectively hereinafter referred to as automated tools, that automatically or semi-automatically send orders for tradable objects to the exchange. Many different trading tools are usually provided to, among other things, facilitate fast and accurate order entry. For instance, an automated or semi-automated tool might quickly calculate one or more order parameters, such as order price or order quantity, based on market conditions, or some other reference condition, and then automatically send an order with these parameters to an exchange for matching.
 - (4) Traders frequently execute trading strategies that require the constant tracking of a large amount of individual tradable objects and the need to assimilate large amounts of data in order to recognize market trends, view current market conditions, and quickly enter, cancel, update, and modify orders. Thus, traders employ specialized trading interfaces for assistance. Trading interfaces present a variety of information, which may include the status of each tradable object in a line of numeric data fields covering key variables relevant to trading, such as price, quantity available at each price, the level of inside market, and others. Traders may regularly operate many such trading interfaces in order to track market conditions for a plurality of tradable objects and execute various trading strategies simultaneously. The use of multiple interfaces can become burdensome to a trader who needs to accurately view critical changes in the market conditions of a large number of tradable objects in order to quickly execute trades. Unless accommodations are made, the interfaces that are supposed to help the trader operate more efficiently and effectively can ultimately become overwhelming and inhibit the trader instead.
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Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) Certain embodiments are disclosed with reference to the following drawings.
- (2) FIG. 1 illustrates a block diagram representative of an example electronic trading system in which certain embodiments may be employed.
- (3) FIG. 2 illustrates a block diagram of another example electronic trading system in which certain embodiments may be employed.
- (4) FIG. 3 illustrates a block diagram of an example computing device which may be used to implement the disclosed embodiments.
- (5) FIGS. 4-9 illustrate an exemplary trading interface for presenting market data relating to a tradable object which may be employed with certain disclosed embodiments and the generation of order entry fields for facilitating the execution of trades of the tradable object, among other things.
- (6) FIGS. 10-12 illustrate a pair of exemplary trading interfaces positioned to illustrate the effects of proximity on the generation of order entry fields for facilitating the execution of trades of the tradable object, among other things.
- (7) FIG. 13 is a flow diagram illustrating an exemplary method for presenting market data and facilitating the execution of trades through dynamically generated displays of order entry fields on a trading interface, among other things, relating to a tradable object which may be employed with certain disclosed embodiments.
- (8) Certain embodiments will be better understood when read in conjunction with the provided figures, which illustrate examples. It should be understood, however, that the embodiments are not limited to the arrangements and instrumentality shown in the attached figures.

DETAILED DESCRIPTION

- (9) The disclosed embodiments generally relate to systems and methods for providing trading interfaces that display market data in a manner that assists traders in executing trades. Some embodiments are directed to systems and methods for displaying market data on a trading interface relating to prices for trading a tradable object which are operatively configured to dynamically generate an order entry field for display on the trading interface in response to detecting or receiving an instance of user input through a data input device within an activation region defined on the trading interface in a positional relationship with market data relating to a unique price for trading the tradable object, wherein the instance of the user input in the activation region generates the display of the order entry field whereby the order entry field facilitates the execution of a trade of the tradable object at the unique price.
- (10) As used herein, the term “tradable object” refers to anything that can be traded with a quantity and/or price. It includes, but is not limited to, all types of traded events, goods and/or financial products, which can include, for example, stocks, options, bonds, futures, currency, and warrants, as well as funds, derivatives and collections of the foregoing, and all types of commodities, such as grains, energy, and metals. The tradable object may be “real,” such as products that are listed by an exchange for trading, or “synthetic,” such as a combination of real products that is created by the user. A tradable object could actually be a combination of other tradable objects, such as a class of tradable objects.
- (11) Although this description discloses embodiments including, among other components, software executed on hardware, it should be noted that the embodiments are merely illustrative and should not be considered as limiting. For example, it is contemplated that any or all of these hardware and software components may be embodied exclusively in hardware, exclusively in software, exclusively in firmware, or in any combination of hardware, software, and/or firmware. Accordingly, certain embodiments may be implemented in other ways. Furthermore, well known process steps and computer components may have not been described in detail herein in order to

avoid unnecessarily obscuring the underlying concepts.

I. Brief Description of Certain Embodiments

(12) In one embodiment, a method for displaying market data and facilitating the execution of trades through a computing device in communication with an electronic exchange is disclosed. The method comprises the steps of displaying a trading interface for presenting current market data

(13) In some embodiments, the trading interface includes a plurality of generally axially aligned value levels defining a central price axis, a plurality of ask regions and a plurality of bid regions. Each of the axially aligned value levels contains a current numerical price for a tradable object in sequential order along the price axis. Each ask region of the plurality of ask regions is transversely aligned to a value level relative to the central price axis and contains a numerical value representing a current ask quantity at the price displayed in the transversely aligned value level, thereby displaying a positional relationship between the current ask quantity and price along the central price axis. Each bid region of the plurality of bid regions is transversely aligned to a value level and contains a numerical value representing a current bid quantity at the price displayed in the transversely aligned value level, thereby displaying a positional relationship between the current bid quantity and price along the central price axis.

(14) In one embodiment, a method as described herein includes the step of generating an order entry field for display on the trading interface in response to receiving an instance of user input within an activation region associated with the order entry field and defined within the trading interface in a positional relationship with a unique axially aligned value level of the plurality of axially aligned value levels, wherein the order entry field facilitates the execution of a trade of the tradable object at the price displayed in the unique axially aligned value level.

(15) In another embodiment, a method for displaying market data and executing trade orders includes the step of displaying a trading interface for presenting current market data in an area of a display device. The trading interface may include value levels, ask regions, bid regions and order entry and associated activation regions defined within the trading interface in a transversely aligned positional relationship with the axially aligned value levels of the plurality of axially aligned value levels. The method of this embodiment may also include the step of generating an order entry field for display on the trading interface in response to positioning a cursor within an activation region, wherein the order entry field facilitates the execution of a trade of the tradable object at the price displayed in the transversely aligned value level.

(16) Some embodiments of a system for displaying market data and executing trade orders which includes a computing device in a communication with an electronic exchange are also disclosed. The system comprises a display device for displaying a trading interface which may include all of the features described above and/or the features of the exemplary embodiments described herein below, and a data input device for generating an instance of user input within an activation region, wherein the computing device is configured to generate an order entry field and the associated activation region for display on the trading interface in response to receiving an instance of user input within an activation region, wherein the order entry field facilitates the execution of a trade of the tradable object at the price displayed in the transversely aligned value level.

(17) Other embodiments are directed to a non-transitory computer-readable information recording medium which stores a program for controlling a computer to receive market data from an electronic exchange, the program comprising code for permitting the computer to perform the steps for displaying the trading interface including all of the features described above and/or the features of the exemplary embodiments described herein below, and generating an order entry field for display on the trading interface in response to receiving an instance of user input within an activation region associated with the order entry field and defined within the trading interface in a positional relationship with a unique axially aligned value level of the plurality of axially aligned value levels, wherein the order entry field facilitates the execution of a trade of the tradable object at the price displayed in the unique axially aligned value level.

II. Example Electronic Trading System

(18) FIG. 1 illustrates a block diagram representative of an example electronic trading system **100** in which certain embodiments may be employed. The system **100** includes a trading device **110**, a gateway **120**, and an exchange **130**. The trading device **110** is in communication with the gateway **120**. The gateway **120** is in communication with the exchange **130**. As used herein, the phrase “in communication” encompasses direct communication and/or indirect communication through one or more intermediary components. The exemplary electronic trading system **100** depicted in FIG. 1 may be in communication with additional components, subsystems, and elements to provide additional functionality and capabilities without departing from the teaching and disclosure provided herein.

(19) In operation, the trading device **110** may receive market data from the exchange **130** through the gateway **120**. A user may utilize the trading device **110** to monitor this market data and/or base a decision to send an order message to buy or sell one or more tradeable objects to the exchange **130**.

(20) Market data may include data about a market for a tradeable object. For example, market data may include the inside market, market depth, last traded price (“LTP”), a last traded quantity (“LTQ”), or a combination thereof. The inside market is the lowest available ask price (best offer) and the highest available bid price (best bid) in the market for a particular tradable object at a particular point in time (since the inside market may vary over time). Market depth refers to quantities available at the inside market and at other prices away from the inside market. Due to the quantity available, there may be “gaps” in market depth.

(21) A tradeable object is anything which may be traded. For example, a certain quantity of the tradeable object may be bought or sold for a particular price. A tradeable object may include, for example, financial products, stocks, options, bonds, future contracts, currency, warrants, funds derivatives, securities, commodities, swaps, interest rate products, index-based products, traded events, goods, or a combination thereof. A tradeable object may include a product listed and/or administered by an exchange (for example, the exchange **130**), a product defined by the user, a combination of real or synthetic products, or a combination thereof. There may be a synthetic tradeable object that corresponds and/or is similar to a real tradeable object.

(22) An order message is a message that includes a trade order. A trade order may be, for example, a command to place an order to buy or sell a tradeable object, a command to initiate managing orders according to a defined trading strategy, a command to change or cancel a previously submitted order (for example, modify a working order), an instruction to an electronic exchange relating to an order, or a combination thereof.

(23) The trading device **110** may include one or more electronic computing platforms. For example, the trading device **110** may include a desktop computer, hand-held device, laptop, server, a portable computing device, a trading terminal, an embedded trading system, a workstation, an algorithmic trading system such as a “black box” or “grey box” system, cluster of computers, or a combination thereof. As another example, the trading device **110** may include a single or multi-core processor in communication with a memory or other storage medium configured to accessibly store one or more computer programs, applications, libraries, computer readable instructions, and the like, for execution by the processor.

(24) As used herein, the phrases “configured to” and “adapted to” encompass that an element, structure, or device has been modified, arranged, changed, or varied to perform a specific function or for a specific purpose.

(25) By way of example, the trading device **110** may be implemented as a personal computer running a copy of X_TRADER®, an electronic trading platform provided by Trading Technologies International, Inc. of Chicago, Illinois (“Trading Technologies”). As another example, the trading device **110** may be a server running a trading application providing automated trading tools such as ADL™, AUTOSPREADER®, and/or AUTOTRADER™, also provided by Trading Technologies.

In yet another example, the trading device **110** may include a trading terminal in communication with a server, where collectively the trading terminal and the server are the trading device **110**.

(26) The trading device **110** is generally owned, operated, controlled, programmed, configured, or otherwise used by a user. As used herein, the phrase “user” may include, but is not limited to, a human (for example, a trader), trading group (for example, group of traders), or an electronic trading device (for example, an algorithmic trading system). One or more users may be involved in the ownership, operation, control, programming, configuration, or other use, for example.

(27) The trading device **110** may include one or more trading applications. As used herein, a trading application is an application that facilitates or improves electronic trading. A trading application provides one or more electronic trading tools. For example, a trading application stored by a trading device may be executed to arrange and display market data in one or more trading windows. In another example, a trading application may include an automated spread trading application providing spread trading tools. In yet another example, a trading application may include an algorithmic trading application that automatically processes an algorithm and performs certain actions, such as placing an order, modifying an existing order, deleting an order. In yet another example, a trading application may provide one or more trading screens. A trading screen may provide one or more trading tools that allow interaction with one or more markets. For example, a trading tool may allow a user to obtain and view market data, set order entry parameters, submit order messages to an exchange, deploy trading algorithms, and/or monitor positions while implementing various trading strategies. The electronic trading tools provided by the trading application may always be available or may be available only in certain configurations or operating modes of the trading application.

(28) A trading application may include computer readable instructions that are stored in a computer readable medium and executable by a processor. A computer readable medium may include various types of volatile and non-volatile storage media, including, for example, random access memory, read-only memory, programmable read-only memory, electrically programmable read-only memory, electrically erasable read-only memory, flash memory, any combination thereof, or any other tangible data storage device. As used herein, the term non-transitory or tangible computer readable medium is expressly defined to include any type of computer readable storage media and to exclude propagating signals.

(29) One or more components or modules of a trading application may be loaded into the computer readable medium of the trading device **110** from another computer readable medium. For example, the trading application (or updates to the trading application) may be stored by a manufacturer, developer, or publisher on one or more CDs or DVDs, which are then loaded onto the trading device **110** or to a server from which the trading device **110** retrieves the trading application. As another example, the trading device **110** may receive the trading application (or updates to the trading application) from a server, for example, via the Internet or an internal network. The trading device **110** may receive the trading application or updates when requested by the trading device **110** (for example, “pull distribution”) and/or un-requested by the trading device **110** (for example, “push distribution”).

(30) The trading device **110** may be adapted to send order messages. For example, the order messages may be sent to through the gateway **120** to the exchange **130**. As another example, the trading device **110** may be adapted to send order messages to a simulated exchange in a simulation environment which does not effectuate real-world trades.

(31) The order messages may be sent at the request of a user. For example, a trader may utilize the trading device **110** to send an order message or manually input one or more parameters for a trade order (for example, an order price and/or quantity). As another example, an automated trading tool provided by a trading application may calculate one or more parameters for a trade order and automatically send the order message. In some instances, an automated trading tool may prepare the order message to be sent but not actually send it without confirmation from a user.

(32) An order message may be sent in one or more data packets or through a shared memory system. For example, an order message may be sent from the trading device **110** to the exchange **130** through the gateway **120**. The trading device **110** may communicate with the gateway **120** using a local area network, a wide area network, a wireless network, a virtual private network, a T1 line, a T3 line, an integrated services digital network (“ISDN”) line, a point-of-presence, the Internet, and/or a shared memory system, for example.

(33) The gateway **120** may include one or more electronic computing platforms. For example, the gateway **120** may implemented as one or more desktop computer, hand-held device, laptop, server, a portable computing device, a trading terminal, an embedded trading system, workstation with a single or multi-core processor, an algorithmic trading system such as a “black box” or “grey box” system, cluster of computers, or any combination thereof.

(34) The gateway **120** may facilitate communication. For example, the gateway **120** may perform protocol translation for data communicated between the trading device **110** and the exchange **130**. The gateway **120** may process an order message received from the trading device **110** into a data format understood by the exchange **130**, for example. Similarly, the gateway **120** may transform market data in an exchange-specific format received from the exchange **130** into a format understood by the trading device **110**, for example.

(35) The gateway **120** may include a trading application, similar to the trading applications discussed above, that facilitates or improves electronic trading. For example, the gateway **120** may include a trading application that tracks orders from the trading device **110** and updates the status of the order based on fill confirmations received from the exchange **130**. As another example, the gateway **120** may include a trading application that coalesces market data from the exchange **130** and provides it to the trading device **110**. In yet another example, the gateway **120** may include a trading application that provides risk processing, calculates implieds, handles order processing, handles market data processing, or a combination thereof.

(36) In certain embodiments, the gateway **120** communicates with the exchange **130** using a local area network, a wide area network, a virtual private network, a T1 line, a T3 line, an ISDN line, a point-of-presence, the Internet, and/or a shared memory system, for example.

(37) The exchange **130** may be owned, operated, controlled, or used by an exchange entity. Example exchange entities include the CME Group, the London International Financial Futures and Options Exchange, the Intercontinental Exchange, and Eurex. The exchange **130** may include an electronic matching system, such as a computer, server, or other computing device, which is adapted to allow tradeable objects, for example, offered for trading by the exchange, to be bought and sold. The exchange **130** may include separate entities, some of which list and/or administer tradeable objects and others which receive and match orders, for example. The exchange **130** may include an electronic communication network (“ECN”), for example.

(38) The exchange **130** may be an electronic exchange. The exchange **130** is adapted to receive order messages and match contra-side trade orders to buy and sell tradeable objects. Unmatched trade orders may be listed for trading by the exchange **130**. The trade orders may include trade orders received from the trading device **110** or other devices in communication with the exchange **130**, for example. For example, typically the exchange **130** will be in communication with a variety of other trading devices (which may be similar to trading device **110**) which also provide trade orders to be matched.

(39) The exchange **130** is adapted to provide market data. Market data may be provided in one or more messages or data packets or through a shared memory system. For example, the exchange **130** may publish a data feed to subscribing devices, such as the trading device **110** or gateway **120**. The data feed may include market data.

(40) The system **100** may include additional, different, or fewer components. For example, the system **100** may include multiple trading devices, gateways, and/or exchanges. In another example, the system **100** may include other communication devices, such as middleware, firewalls, hubs,

switches, routers, servers, exchange-specific communication equipment, modems, security managers, and/or encryption/decryption devices.

III. Expanded Example Electronic Trading System

(41) FIG. 2 illustrates a block diagram of another example electronic trading system **200** in which certain embodiments may be employed. In this example, a trading device **210a** is in communication with an exchange **230a** through a gateway **220a**. The following discussion mainly focuses on the trading device **210a**, gateway **220a**, and the exchange **230a**. However, the trading device **210a** may also be connected to and communicate with any number of gateways **220n** connected to exchanges **230n**. The communication between the trading device **110a** and other exchanges **230n** may be the same, similar, or different than the communication between the trading device **210a** and exchange **230a**. Generally, each exchange has its own preferred techniques and/or formats for communicating with a trading device, a gateway, the user, or another exchange.

(42) The trading device **210a**, which may be similar to the trading device **110** in FIG. 1, may include a server **212a** in communication with a trading terminal **214a**. The server **212a** may be located geographically closer to the gateway **120** than the trading terminal **214a**. As a result, the server **212a** latency benefits that are not afforded to the trading terminal **214a**. In operation, the trading terminal **214a** may provide a trading screen to a user and communicate commands to the server **212a** for further processing. For example, a trading algorithm may be deployed to the server **212a** for execution based on market data. The server **212a** may execute the trading algorithm without further input from the user. In another example, the server **212a** may include a trading application providing automated trading tools and communicate back to the trading terminal **214a**. The trading device **210a** may include, additional, different, or fewer components.

(43) The trading device **210a** may communicate with the gateway **220a** using one or more communication networks. As used herein, a communication network is any network, including the Internet, which facilitates or enables communication between, for example, the trading device **210a**, the gateway **220a** and the exchange **220a**. For example, as shown in FIG. 2, the trading device **210a** may communicate with the gateway **220a** across a multicast communication network **202a**. The data on the network **202a** may be logically separated by subject (for example, prices, orders, or fills). As a result, the server **212a** and trading terminal **214a** can subscribe to and receive data (for example, data relating to prices, orders, or fills) depending on their individual needs.

(44) The gateway **220a**, which may be similar to the gateway **120** of FIG. 1, may include a price server **222a**, order server **224a**, and fill server **226a**. The gateway **220a** may include additional, different, or fewer components. The price server **222a** may process price data. Price data includes data related to a market for one or more tradeable objects. The order server **224a** may process order data. Order data is data related to a user's trade orders. For example, order data may include order messages, confirmation messages, or other types of messages. The fill server collects and provides fill data. Fill data includes data relating to one or more fills of trade orders. For example, the fill server **226a** may provide a record of trade orders, which have been routed through the order server **224a**, that have and have not been filled. The servers **222a**, **224a**, **226a** may run on the same machine or separate machines.

(45) The gateway **220a** may communicate with the exchange **230a** using one or more communication networks. For example, as shown in FIG. 2, there may be two communication networks connecting the gateway **220a** and the exchange **230a**. The network **204a** may be used to communicate market data to the price server **222a**. In some instances, the exchange **230a** may include this data in a data feed that is published to subscribing devices. The network **206a** may be used to communicate order data.

(46) The exchange **230a**, which may be similar to the exchange **130** of FIG. 1, may include an order book **232a** and a matching engine **234a**. The exchange **230a** may include additional, different, or fewer components. The order book **232a** is a database that includes data relating to unmatched quantity of trade orders. For example, an order book may include data relating to a

market for a tradeable object, such as the inside market, market depth at various price levels, the last traded price, and the last traded quantity. The matching engine **234a** may match contra-side bids and offers. For example, the matching engine **234a** may execute one or more matching algorithms that match contra-side bids and offers. A sell order is contra-side to a buy order with the same price. Similarly, a buy order is contra-side to a sell order with the same price.

(47) In operation, the exchange **230a** may provide price data from the order book **232a** to the price server **222a** and order data and/or fill data from the matching engine **234a** to the order server **224a**. Servers **222a**, **224a**, **226a** may translate and communicate this data back to the trading device **210a**. The trading device **210a**, for example, using a trading application, may process this data. For example, the data may be displayed to a user. In another example, the data may be utilized in a trading algorithm to determine whether a trade order should be submitted to the exchange **230a**. The trading device **210a** may prepare and send an order message to the exchange **230a**.

(48) In certain embodiments, the gateway **220a** is part of the trading device **210a**. For example, the components of the gateway **220a** may be part of the same computing platform as the trading device **210a**. As another example, the functionality of the gateway **220a** may be performed by components of the trading device **210a**. In certain embodiments, the gateway **220a** is not present. Such an arrangement may occur when the trading device **210a** does not need to utilize the gateway **220a** to communicate with the exchange **230a**, for example. For example, if the trading device **210a** has been adapted to communicate directly with the exchange **230a**.

(49) Additional trading devices **210b-210e**, which are similar to trading device **210a**, may be connected to one or more of the gateways **220a-220n** and exchanges **230a-230n**. Furthermore, additional gateways **220n**, similar to the gateway **220a**, include a price server **222n**, an order server **224n**, and a fill server **226n**. The gateway **220n** may be in communication with multiple exchanges **230n**, similar to the exchange **230a** utilizing, for example, networks **204n**, and **206n**. The exchange **230n** may include an order book **232n** and a match engine **234n**. Each gateway may be in communication with one or more different exchanges, for example. Such an arrangement may, for example, allow one or more trading devices **210a** to trade at more than one exchange (and/or provide redundant connections to multiple exchanges) utilizing, for example, networks **202a**, **202n**.

IV. Example Computing Device

(50) FIG. 3 illustrates a block diagram of an example computing device **300** which may be used to implement the disclosed embodiments. The trading device **110** of FIG. 1 may include one or more computing devices **300**, for example. The gateway **120** of FIG. 1 may include one or more computing devices **300**, for example. The exchange **130** of FIG. 1 may include one or more computing devices **300**, for example.

(51) The computing device **300** includes a communication network **310**, a processor **312**, a memory **314**, an interface **316**, an input device **318**, and an output device **320**. The computing device **300** may include additional, different, or fewer components. For example, multiple communication networks, multiple processors, multiple memory, multiple interfaces, multiple input devices, multiple output devices, or any combination thereof, may be provided. As another example, the computing device **300** may not include an input device **318** or output device **320**.

(52) As shown in FIG. 3, the computing device **300** may include a processor **312** coupled to a communication network **310**. The communication network **310** may include a communication bus, channel, electrical or optical network, circuit, switch, fabric, or other mechanism for communicating data between components in the computing device **300**. The communication network **310** may be communicatively coupled with and transfer data between any of the components of the computing device **300**.

(53) The processor **312** may be any suitable processor, processing unit, or microprocessor. The processor **312** may include one or more general processors, digital signal processors, application specific integrated circuits, field programmable gate arrays, analog circuits, digital circuits, programmed processors, and/or combinations thereof, for example. The processor **312** may be a

single device or a combination of devices, such as one or more devices associated with a network or distributed processing. Any processing strategy may be used, such as multi-processing, multi-tasking, parallel processing, and/or remote processing. Processing may be local or remote and may be moved from one processor to another processor. In certain embodiments, the computing device **300** is a multi-processor system and, thus, may include one or more additional processors which are communicatively coupled to the communication network **310**.

(54) The processor **312** may be operable to execute logic and other computer readable instructions encoded in one or more tangible media, such as the memory **314**. As used herein, logic encoded in one or more tangible media includes instructions which may be executable by the processor **312** or a different processor. The logic may be stored as part of software, hardware, integrated circuits, firmware, and/or micro-code, for example. The logic may be received from an external communication device via a communication network such as the network **340**. The processor **312** may execute the logic to perform the functions, acts, or tasks illustrated in the figures or described herein.

(55) The memory **314** may be one or more tangible media, such as computer readable storage media, for example. Computer readable storage media may include various types of volatile and non-volatile storage media, including, for example, random access memory, read-only memory, programmable read-only memory, electrically programmable read-only memory, electrically erasable read-only memory, flash memory, any combination thereof, or any other tangible data storage device. As used herein, the term non-transitory or tangible computer readable medium is expressly defined to include any type of computer readable medium and to exclude propagating signals. The memory **314** may include any desired type of mass storage device including hard disk drives, optical media, magnetic tape or disk, etc.

(56) The memory **314** may include one or more memory devices. For example, the memory **314** may include local memory, a mass storage device, volatile memory, non-volatile memory, or a combination thereof. The memory **314** may be adjacent to, part of, programmed with, networked with, and/or remote from processor **312**, so the data stored in the memory **314** may be retrieved and processed by the processor **312**, for example. The memory **314** may store instructions which are executable by the processor **312**. The instructions may be executed to perform one or more of the acts or functions described herein or shown in the figures.

(57) The memory **314** may store a trading application **330**. In certain embodiments, the trading application **330** may be accessed from or stored in different locations. The processor **312** may access the trading application **330** stored in the memory **314** and execute computer-readable instructions included in the trading application **330**.

(58) In certain embodiments, during an installation process, the trading application may be transferred from the input device **318** and/or the network **340** to the memory **314**. When the computing device **300** is running or preparing to run the trading application **330**, the processor **312** may retrieve the instructions from the memory **314** via the communication network **310**.

V. Dynamically Generating Order Entry Fields

(59) Exemplary embodiments described below include a trading interface that presents market data and dynamically generates order entry fields in response to user input which are intended to allow a trader to gain an accurate and comprehensive view of the current market conditions for a large number of tradable objects and execute trades quickly, among other things. The dynamically generated order entry fields provide a mechanism by which a user may enter an order via a trading interface configured and minimized to reduce the amount of display area necessary to utilize the displayed market data. The reduced footprint of the example trading interface configured allows the trader to display more information without having to sacrifice functionality such as the ability to enter trades efficiently. As used herein, the term “market data” is not intended to be limited to data and information describing trade orders and market conditions, and may include any information received directly or indirectly from the exchange **130** or data generated by the trading device **300**.

Embodiments discussed herein describe the user input as involving movement of a cursor about the trading interface which may be directed via a data input device such as a mouse, keyboard, touchpad or any alternative data input devices or means.

(60) FIG. 4 provides an embodiment of an exemplary trading interface of the invention for presenting market data relating to a tradable object that is referred to herein as a trading interface **402**. Interface **402** may be displayed in a display area **400** defined on any type of display device in communication with computing device **300**, and may be connected with or otherwise comprise output device **320**.

(61) In this embodiment, trading interface **402** includes a plurality of value levels **404** arranged in axial alignment with one another defining a value axis **406**. Interface **402** further includes a plurality of bid quantity regions **408** and a plurality of ask quantity regions **410**. The plurality of bid quantity regions **408** are axially aligned with one another defining a bid quantity axis **412** having a centerline or axis aligned in parallel to value axis **406**. The plurality of bid quantity regions **408** are located in a generally a transverse position with respect to value axis **406** defined by the plurality of value levels **404**. The plurality of ask quantity regions **410** are axially aligned with one another defining an ask quantity axis **414** having a centerline or axis aligned in parallel to value axis **406**. The plurality of ask quantity regions **410** are therefore located in a generally a transverse position with respect to value axis **406** defined by the plurality of value levels **404**, similarly to the plurality of bid quantity regions **408** but on the opposing side of value axis **406** in this embodiment. In one embodiment, the value axis **406** along with the bid quantity axis **412** and ask quantity axis **414** are aligned and arranged to define a ladder trading interface or ladder interface.

(62) Each value level **404** in interface **402** contains a numerical value representing a price for the tradable object. Either the bid quantity region **408** or the ask quantity region **410** transversely adjacent to each value level **404** may contain a numerical value representing the bid quantity or ask quantity for the tradable object at the numerical price shown in the corresponding value level **404**. In this embodiment, the prices shown in value levels **404** are presented in sequential order from lowest price at a bottom **416** of interface **402** to highest price at a top **418** of interface **402**.

(63) FIGS. 5-7 illustrate the generation of an order entry field **420A** for display in an area adjacent to a value level **404A** of trading interface **402**. In this embodiment, the generation of order entry fields such as field **420A** shown in FIG. 5 is responsive to movement of a cursor "C" as directed by a user through a data input device into an activation region **422A** associated with the order entry field defined on interface **402** as shown particularly in FIG. 6.

(64) In this embodiment, activation regions are not visible on interface **402** and are defined adjacent to each value level **404** such that movement of a cursor towards a value level **404** will result in the cursor being positioned within an activation region and generate the display of an order entry field. Each activation region is associated with a specific value level **404** and generates an order entry field whereby entering an order amount in the field facilitates the execution of a trade of the tradable object at the price shown in the specific value level **404** associated with the activation region. It should be understood that the relative size and position of the activation regions may vary. However, as exemplified in FIG. 7, each activation region **422** of this embodiment is in a transversely aligned positional relationship perpendicular to the value axis **406** with respect to its associated value level **404** and may extend transversely over a portion or all bid quantity regions **408** and/or ask quantity regions **410** as well as outside of trading interface **402**. Order entry fields **420** may be generated and displayed within their respective activation regions **422**. For example, when the cursor C enters the activation region **422A** shown in FIG. 6, an order entry field **420A** (see FIG. 5) may be generated and displayed. Alternatively, when the cursor C enters the activation region **422A** shown in FIG. 6, an order entry field **420A** (see FIG. 5) may be generated and displayed elsewhere on trading interface **402**, display area **400**. In other embodiments, adjacent activation regions may overlap and the movement of the cursor C within those regions and/or the projected or calculated trajectory of the cursor C may be utilized to predict

which order entry field **420A** should be activated.

(65) For example, entering a number *X* into field **420A** would facilitate the execution of a trade of *X* amount of the tradable object at the price shown in value level **404A**, that is, “176.” It should be understood that the prices in value levels **404** may change as market data fluctuates and is presented in trading interface **402**.

(66) FIGS. **8** and **9** illustrate the generation of an order entry field **420B** in FIG. **9** as a result of positioning the cursor *C* within an activation region **422B** in FIG. **8** associated with value level **404B**. As discussed above, entering a number *X* into order entry field **420B** in FIG. **9** will facilitate the execution of a trade of *X* amount of the tradable object at the price shown in value level **404B** (i.e., “178”).

(67) It should be understood that trading interfaces of the invention, such as interface **402** may incorporate user-friendly features and fit seamlessly with other operating system interfaces, that is, in a framed form having borders, multiple folders, toolbars with pull-down menus, embedded links to other screens and various other selectable features associated with animated graphical representations of depressible buttons that may be can be selected (i.e., “clicked on” or “touched”) by the user via input device **318**, which may include or comprise a connected mouse, keyboard or touch-enabled display device.

(68) FIGS. **10-12** illustrate a pair of exemplary trading interfaces **402** and **402'** generated and displayed in various adjacent positions with respect to one another on a display device in communication with computing device **300** to show the effects of proximity on the size and position of the activation regions **422A** and **422B** generated by each interface **402** and **402'**, respectively. FIG. **10** illustrates a pair of trading interfaces **402** and **402'** generated and displayed via the display area **400**. The example trading interfaces **402** and **402'** are positioned a distance *X* from each other in display area **400** such that the activation region **422B** generated by the interface **402** does not interfere with or otherwise influence the definition and creation of another activation region **422A** by the interface **402'**.

(69) FIG. **11** illustrates the pair of trading interfaces **402** and **402'** from FIG. **10** positioned a distance *Y* from each other in display area **400**. In the illustrated example, the distance *X* is greater than the distance *Y*. In other words, the distance between the trading interface **402** and **402'** is greater in the illustration shown in FIG. **10** than it is in the illustration shown in FIG. **11**. In certain embodiments, the proximity of the two trading interfaces **402** and **402'** affects the size and/or shape of the corresponding activation regions **422B** and **422A** (see FIG. **10**). For example, when the trading interface **402** is positioned in close proximity to the trading interface **402'** in display area **400** as shown in FIG. **11**, the activation regions **422D** and **422C** automatically resize to the extent necessary to ensure that the regions do not crossover or otherwise interfere with each other. FIG. **12** illustrates an embodiment where the trading interface **402** is positioned substantially adjacent (i.e., the distance between the two interfaces is negligible, zero or near zero) to the trading interface **402'**. In the exemplary arrangement of FIG. **12**, the activation region **422F** and the activation region **422E** have been resized and positioned within the respective value levels and quantity regions of the trading interface **402** and **402'**.

(70) FIG. **13** provides exemplary functional steps of a method of carrying out the invention generally referred to as method **550** through a computing device such as computing device **300**.

(71) As shown by step **552**, market data for a tradable object is received by, for example, computing device **300**, through one or more sources. The most current market data is displayed in step **554** on a display device through a trading interface that formats the market data, such as trading interface **402**, which may include displaying prices for the tradable object in value levels, along with bid quantities or ask quantities for the displayed prices.

(72) In step **556**, activation regions are defined on the trading interface in a positional relationship with the value levels containing prices for the tradable object, with each activation region being associated with a unique value level containing a price for the tradable object as provided by the

most current market data. As shown by steps 558 and 560, an order entry field is generated for display responsive to detecting the cursor or other user input within an activation region. The order entry field generated facilitates execution of a trade at the unique price displayed in the value level associated with the activation region in which the cursor is positioned.

(73) Some of the described figures depict example block diagrams, systems, and/or flow diagrams representative of methods that may be used to implement all or part of certain embodiments. One or more of the components, elements, blocks, and/or functionality of the example block diagrams, systems, and/or flow diagrams may be implemented alone or in combination in hardware, firmware, discrete logic, as a set of computer readable instructions stored on a tangible computer readable medium, and/or any combinations thereof, for example.

(74) The example block diagrams, systems, and/or flow diagrams may be implemented using any combination of application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)), field programmable logic device(s) (FPLD(s)), discrete logic, hardware, and/or firmware, for example. Also, some or all of the example methods may be implemented manually or in combination with the foregoing techniques, for example.

(75) The example block diagrams, systems, and/or flow diagrams may be performed using one or more processors, controllers, and/or other processing devices, for example. For example, the examples may be implemented using coded instructions, for example, computer readable instructions, stored on a tangible computer readable medium. A tangible computer readable medium may include various types of volatile and non-volatile storage media, including, for example, random access memory (RAM), read-only memory (ROM), programmable read-only memory (PROM), electrically programmable read-only memory (EPROM), electrically erasable read-only memory (EEPROM), flash memory, a hard disk drive, optical media, magnetic tape, a file server, any other tangible data storage device, or any combination thereof. The tangible computer readable medium is non-transitory.

(76) Further, although the example block diagrams, systems, and/or flow diagrams are described above with reference to the figures, other implementations may be employed. For example, the order of execution of the components, elements, blocks, and/or functionality may be changed and/or some of the components, elements, blocks, and/or functionality described may be changed, eliminated, sub-divided, or combined. Additionally, any or all of the components, elements, blocks, and/or functionality may be performed sequentially and/or in parallel by, for example, separate processing threads, processors, devices, discrete logic, and/or circuits.

(77) While embodiments have been disclosed, various changes may be made and equivalents may be substituted. In addition, many modifications may be made to adapt a particular situation or material. Therefore, it is intended that the disclosed technology not be limited to the particular embodiments disclosed, but will include all embodiments falling within the scope of the appended claims.

Claims

1. A method including: displaying, by a computing device including a trading application, the computing device in communication with an electronic exchange, a first trading interface of the trading application including (i) a first plurality of value levels aligned along a first value axis, each value level of the first plurality of value levels associated with a price for a first tradeable object, and (ii) a first plurality of bid regions aligned along the first value axis and a first plurality of ask regions aligned along the first value axis, each bid region in the first plurality of bid regions associated with a value level of the first value axis and each ask region in the first plurality of ask regions associated with a value level of the first value axis; displaying, by the computing device, a second trading interface of the trading application including (i) a second plurality of value levels aligned along a second value axis, each value level of the second plurality of value levels

associated with a price for a second tradeable object, and (ii) a second plurality of bid regions aligned along the second value axis and a second plurality of ask regions aligned along the second value axis, each bid region in the second plurality of bid regions associated with a value level of the second value axis and each ask region in the second plurality of ask regions associated with a value level of the second value axis; generating, by the computing device, a first activation region associated with a first value level of the first value axis of the first trading interface of the trading application and aligned with the first value axis at the corresponding first value level, the first activation region extending outside the first trading interface; generating, by the computing device, a second activation region associated with a second value level of the second value axis of the second trading interface of the trading application and aligned with the second value axis at the corresponding second value level, the second activation region extending outside the second trading interface; receiving, by the computing device, a command from a user moving the first trading interface to a proximity with respect to the second trading interface; and automatically resizing, by the computing device, on the proximity, the first activation region and the second activation region.

2. The method of claim 1, wherein the first activation region overlaps at least a portion of one of: (i) a first bid region in the first plurality of bid regions and (ii) a first ask region in the first plurality of ask regions, wherein the second activation region overlaps at least a portion of one of: (i) a second bid region in the second plurality of bid regions and (ii) a second ask region in the second plurality of ask regions.

3. The method of claim 1, further including: generating a first order entry field when an instance of user input is received within the first activation region; displaying, in response to generating the first order entry field, the first order entry field, wherein the first order entry field is displayed adjacent to one of: (i) a first bid region in the first plurality of bid regions associated with the first value level and (ii) a first ask region in the first plurality of ask regions associated with the first value level; generating a second order entry field when an instance of user input is received within the second activation region; and displaying, in response to generating the second order entry field, the second order entry field, wherein the second order entry field is displayed adjacent to one of: (i) a second bid region in the second plurality of bid regions associated with the second value level and (ii) a second ask region in the second plurality of ask regions associated with the second value level.

4. The method of claim 3, further including: receiving by the computing device a first quantity value from the user in the displayed first order entry field; and sending by the computing device an order for the first tradeable object at the price corresponding to the first value level and for the first received quantity value.

5. The method of claim 3, further including: receiving by the computing device a second quantity value from the user in the displayed second order entry field; and sending by the computing device an order for the second tradeable object at the price corresponding to the second value level and for the second received quantity value.

6. The method of claim 1, wherein the automatic resizing includes at least one of: reducing a size of the first activation region outside the first trading interface or reducing a size of the second activation region outside the second trading interface.

7. The method of claim 1, wherein the automatic resizing includes at least one of: reducing a size of the first activation region to not extend outside the first trading interface or reducing a size of the second activation region to not extend outside the second trading interface.

8. A non-transitory computer readable medium having stored therein instructions that when executed by a processor perform a set of operations including: displaying, by a computing device including a trading application, the computing device in communication with an electronic exchange, a first trading interface of the trading application including (i) a first plurality of value levels aligned along a first value axis, each value level of the first plurality of value levels

associated with a price for a first tradeable object, and (ii) a first plurality of bid regions aligned along the first value axis and a first plurality of ask regions aligned along the first value axis, each bid region in the first plurality of bid regions associated with a value level of the first value axis and each ask region in the first plurality of ask regions associated with a value level of the first value axis; displaying, by the computing device, a second trading interface of the trading application including (i) a second plurality of value levels aligned along a second value axis, each value level of the second plurality of value levels associated with a price for a second tradeable object, and (ii) a second plurality of bid regions aligned along the second value axis and a second plurality of ask regions aligned along the second value axis, each bid region in the second plurality of bid regions associated with a value level of the second value axis and each ask region in the second plurality of ask regions associated with a value level of the second value axis; generating, by the computing device, a first activation region associated with a first value level of the first value axis of the first trading interface of the trading application and aligned with the first value axis at the corresponding first value level, the first activation region extending outside the first trading interface; generating, by the computing device, a second activation region associated with a second value level of the second value axis of the second trading interface of the trading application and aligned with the second value axis at the corresponding second value level, the second activation region extending outside the second trading interface; receiving, by the computing device, a command from a user moving the first trading interface to a proximity with respect to the second trading interface; and automatically resizing, by the computing device, based on the proximity, the first activation region and the second activation region.

9. The non-transitory computer readable medium of claim 8, wherein the first activation region overlaps at least a portion of one of: (i) a first bid region in the first plurality of bid regions and (ii) a first ask region in the first plurality of ask regions, wherein the second activation region overlaps at least a portion of one of: (i) a second bid region in the second plurality of bid regions and (ii) a second ask region in the second plurality of ask regions.

10. The non-transitory computer readable medium of claim 8, wherein the set of operations further includes: generating a first order entry field when an instance of user input is received within the first activation region; displaying, in response to generating the first order entry field, the first order entry field, wherein the first order entry field is displayed adjacent to one of: (i) a first bid region in the first plurality of bid regions associated with the first value level and (ii) a first ask region in the first plurality of ask regions associated with the first value level; generating a second order entry field when an instance of user input is received within the second activation region; and displaying, in response to generating the second order entry field, the second order entry field, wherein the second order entry field is displayed adjacent to one of: (i) a second bid region in the second plurality of bid regions associated with the second value level and (ii) a second ask region in the second plurality of ask regions associated with the second value level.

11. The non-transitory computer readable medium of claim 10, wherein the set of operations further include: receiving, by the computing device, a first quantity value from the user in the displayed first order entry field; and sending, by the computing device, an order for the first tradeable object at the price corresponding to the first value level and for the first received quantity value.

12. The non-transitory computer readable medium of claim 10, wherein the set of operations further include: receiving, by the computing device, a second quantity value from the user in the displayed second order entry field; and sending, by the computing device, an order for the second tradeable object at the price corresponding to the second value level and for the second received quantity value.

13. The non-transitory computer readable medium of claim 8, wherein the automatic resizing includes at least one of: reducing a size of the first activation region outside the first trading interface or reducing a size of the second activation region outside the second trading interface.

14. The non-transitory computer readable medium of claim 8, wherein the automatic resizing

includes at least one of: reducing a size of the first activation region to not extend outside the first trading interface or reducing a size of the second activation region to not extend outside the second trading interface.

15. A system including: a computing device including a trading application, the computing device in communication with an electronic exchange, wherein the computing device is configured to execute instructions to perform a set of operations including: displaying, by the computing device, a first trading interface of the trading application including (i) a first plurality of value levels aligned along a first value axis, each value level of the first plurality of value levels associated with a price for a first tradeable object, and (ii) a first plurality of bid regions aligned along the first value axis and a first plurality of ask regions aligned along the first value axis, each bid region in the first plurality of bid regions associated with a value level of the first value axis and each ask region in the first plurality of ask regions associated with a value level of the first value axis; displaying, by the computing device, a second trading interface of the trading application including (i) a second plurality of value levels aligned along a second value axis, each value level of the second plurality of value levels associated with a price for a second tradeable object, and (ii) a second plurality of bid regions aligned along the second value axis and a second plurality of ask regions aligned along the second value axis, each bid region in the second plurality of bid regions associated with a value level of the second value axis and each ask region in the second plurality of ask regions associated with a value level of the second value axis; generating, by the computing device, a first activation region associated with a first value level of the first value axis of the first trading interface; generating, by the computing device, a second activation region associated with a second value level of the second value axis of the second trading interface of the trading application and aligned with the second value axis at the corresponding second value level, the second activation region extending outside the second trading interface; receiving, by the computing device, a command from a user moving the first trading interface to a proximity with respect to the second trading interface; and automatically resizing, by the computing device, based on the proximity the first activation region and the second activation region to ensure that the first activation region and the second activation region.

16. The system of claim 15, wherein the set of operations further includes: generating a first order entry field when an instance of user input is received within the first activation region; displaying, in response to generating the first order entry field, the first order entry field, wherein the first order entry field is displayed adjacent to one of: (i) a first bid region in the first plurality of bid regions associated with the first value level and (ii) a first ask region in the first plurality of ask regions associated with the first value level; generating a second order entry field when an instance of user input is received within the second activation region; and displaying, in response to generating the second order entry field, the second order entry field, wherein the second order entry field is displayed adjacent to one of: (i) a second bid region in the second plurality of bid regions associated with the second value level and (ii) a second ask region in the second plurality of ask regions associated with the second value level.

17. The system of claim 16, wherein the set of operations further include: receiving, by the computing device, a first quantity value from the user in the displayed first order entry field; and sending, by the computing device, an order for the first tradeable object at the price corresponding to the first value level and for the first received quantity value.

18. The system of claim 16, wherein the set of operations further include: receiving, by the computing device, a second quantity value from the user in the displayed second order entry field; and sending, by the computing device, an order for the second tradeable object at the price corresponding to the second value level and for the second received quantity value.

19. The system of claim 15, wherein the automatic resizing includes at least one of: reducing a size of the first activation region outside the first trading interface or reducing a size of the second activation region outside the second trading interface.

20. The system of claim 15, wherein the automatic resizing includes at least one of: reducing a size of the first activation region to not extend outside the first trading interface or reducing a size of the second activation region to not extend outside the second trading interface.
